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Air nanobubble effectively enhances tetracycline degradation and antibiotic resistance genes reduction in sludge anaerobic digestion

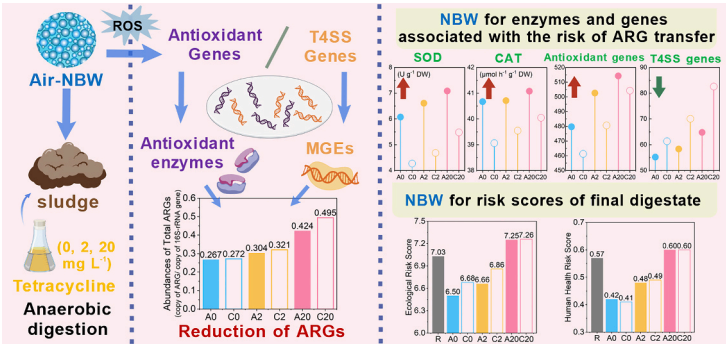
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HIGHLIGHTS

- The oxidative radicals generated by NBW promoted the degradation of tetracycline.
- NBW promoted antioxidant enzyme activity and reduced ATP synthase activity.
- NBW suppressed the expression of biofilm formation and T4SS functional genes.
- Inhibition of horizontal transfer by NBW resulted in a reduction in ARG abundance.
- NBW reduced ecological risks of final digestate under different tetracycline stress.

GRAPHICAL ABSTRACT



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ABSTRACT

Tetracycline is widely present in sludge, and its residual concentration can vary substantially under different contamination scenarios. Nanobubble water (NBW) has been shown to promote sludge anaerobic digestion (AD), but its effects on tetracycline removal and changes in antibiotic resistance genes (ARGs) during sludge AD remain unclear. This study evaluated the effects of NBW on sludge AD under different tetracycline concentrations (0, 2, and 20 mg L^{-1}). The addition of NBW reduced tetracycline concentrations by 10.4%–23.1% and ARG abundance by 1.8%-14.3%, while mitigating the potential risks associated with ARG transmission. In addition, NBW may enhance antioxidant enzyme activities by upregulating the abundance of antioxidant-related functional genes, thereby strengthening the antioxidant capacity of the system. At the same time, NBW reduced the abundance of type IV secretion system-related functional genes, suggesting a reduced dissemination potential of ARGs associated with mobile genetic elements (MGEs). These changes may contribute to the reduction of ARGs during sludge AD. Overall, NBW improved the removal efficiency of tetracycline and ARGs during sludge AD, providing novel perspectives for mitigating the potential environmental risks associated with the final digestate.

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1. Introduction

Tetracycline ranks among the most extensively utilized antibiotics globally in aquaculture sectors, human medicine, and animal husbandry (Wang et al., 2025b). Over 50% of tetracycline cannot be fully metabolized during usage, resulting in the frequent detection of significant amounts of tetracycline in industrial wastewater, medical wastewater, and livestock manure (Xu et al., 2021). During wastewater treatment, tetracycline tends to be adsorbed onto sludge due to its high affinity for solid phases (Zhao et al., 2020), thus leading to its accumulation in the sludge. However, the levels of tetracycline residues in sludge vary greatly. A multi-city survey of sludge found that the total tetracycline concentrations could reach 0.6–37.9 mg kg^{-1} dry weight, while in some pharmaceutical wastewater treatment plant dewatered sludge, tetracycline concentrations could be as high as 300–2790 mg kg^{-1} dry weight (Cheng et al., 2014; Hou et al., 2016; McClellan and Halden, 2010). The treatment of tetracycline-containing sludge is both highly important and challenging.

As a typical selective pressure, tetracycline in the sludge can drive the generation and enrichment of antibiotic resistance genes (ARGs)-particularly tetracycline resistance genes, which are the most prevalent and diverse ARG subsets associated with tetracycline exposure (Wang et al., 2025b). As emerging environmental pollutants, ARGs can be horizontally transferred between microorganisms via mobile genetic elements (MGEs) such as plasmids and transposons (Zhang et al., 2026). The residual tetracycline exhibits a positive correlation with ARGs abundance, which not only reduces the efficacy of subsequent antibiotic treatment but also poses a persistent threat to ecological stability and human health via the dissemination of ARG-carrying microorganisms across diverse environmental media (Bai et al., 2019; Liu et al., 2023). Consequently, the mitigation of tetracycline-induced ARG pollution has emerged as a critical priority in environmental remediation and public health protection (Peng et al., 2025).

Anaerobic digestion (AD) is a promising sludge biodegradation technology that not only enables biomass resource recovery but also can be used to degrade tetracycline and reduce ARGs abundance (Ding et al., 2025; Wang et al., 2025b; Yang et al., 2022). Notably, tetracycline in sludge may induce microbial cell damage, impair the physiological functions and metabolic activities of microorganisms, reduce digestion efficiency and system stability, thereby resulting in limited removal of ARGs during sludge AD (Pan et al., 2019; Shen et al., 2020; Tang et al., 2023).

Suitable additives can improve tetracycline degradation and intensify the removal of ARGs in AD. The addition of biochar enhanced the tetracycline degradation rate during AD of wastewater by 34.8% compared with the control (Ding et al., 2025). Biochar decreased total ARG abundance during sludge AD by 62.2% (Li et al., 2026). Nano-zero-valent iron treatment increased the tetracycline degradation rate by approximately 4.0% during AD of wastewater (Pan et al., 2019). In addition, nano-zero-valent iron significantly accelerated the removal of tetracycline resistance genes and sulfonamide resistance genes during sludge AD (Pang et al., 2025). These studies indicate that both biochar and nano-zero-valent iron can effectively enhance AD performance and mitigate antibiotic-related risks. However, their practical applications may still be constrained by the additional costs associated with material preparation, modification, dosage, and post-treatment management; Moreover, the introduction of solid additives or reactive nanomaterials may increase the complexity of digestate disposal and pose potential secondary environmental risks, thereby limiting the agricultural utilization of digestate (Chen et al., 2023; Pang et al., 2025). Therefore, it remains necessary to explore alternative additive strategies that combine high efficiency, operational simplicity, and environmental compatibility.

Nanobubble water (NBW) refers to an aqueous solution containing nanobubbles with diameters less than 1000 nm, which exhibits advantages such as high adsorption capacity, enhanced air solubility, and prolonged storage stability compared with deionized water (Pei et al., 2024). Air-NBW can create a microaerophilic environment for facultative microorganisms, facilitating substrate degradation and thereby enhancing hydrolysis efficiency (Hou et al., 2024). NBW can improve the microbial community structure by stimulating enzyme activity and enhancing electron transfer, consequently increasing methane production (Fan et al., 2021; Zhu et al., 2024). Existing studies on NBW in AD systems have mainly focused on improving digestion performance, such as hydrolysis efficiency, substrate conversion, and methane production, whereas its potential role in regulating ARG behavior and dissemination risk has rarely been systematically investigated. Compared with solid additives such as biochar and nano-zero-valent iron, NBW, as a liquid-phase additive strategy, does not introduce additional solid materials into the sludge matrix and exhibits better environmental compatibility (Chen et al., 2023; Hou et al., 2024; Pang et al., 2025). Therefore, NBW may represent a promising alternative to conventional solid additives with potentially lower residue-related risks.

The hydroxyl radicals and superoxide radicals generated during the collapse of nanobubbles possess strong oxidizing properties, which facilitate the oxidative degradation of tetracycline in wastewater (Koundle et al., 2024). However, wastewater is predominantly a liquid-phase system, whereas sludge is a complex solid–liquid two-phase system containing a large number of microbial flocs, organic particles, and inorganic impurities. Tetracycline binds tightly to sludge particles via adsorption, complexation, and other interactions, and consequently, compared with tetracycline in wastewater, tetracycline in sludge may exhibit lower migration and transformation efficiencies as well as higher degradation difficulty (Zhao et al., 2020). Owing to the unique properties, nanobubbles can penetrate into the interior of sludge flocs, which is conducive to enhancing the contact probability between strong oxidative radicals and tetracycline (Hou et al., 2024; Pei et al., 2024). Nanobubbles enter the sludge using water as a carrier, which can improve the mass transfer efficiency of the sludge and is beneficial to optimizing the living environment of functional microorganisms (Chuenchart et al., 2021). All these factors may provide potential feasibility for NBW to exert its effects in the sludge system. It is worth noting that the effect of NBW on tetracycline in the AD system remains unclear. If the NBW can effectively promote the degradation of tetracycline in sludge AD, it may alleviate the selective pressure exerted by tetracycline on microorganisms, thereby potentially creating a favorable environment for the reduction of ARGs.

Therefore, unlike previous NBW-related studies that mainly emphasized digestion performance, the present study specifically focuses on the degradation of tetracycline and the behavior of ARGs of sludge AD under tetracycline stress. It explores the effect of Air-NBW on the removal of tetracycline, tetracycline resistance genes, and total ARGs, as well as the potential environmental risks of ARGs. Changes in antioxidant enzyme activities and functional gene expression related to ARG dissemination were explored, aiming to further elucidate the pathways by which NBW addition alters ARGs.

2. Materials and methods

2.1. Materials

The thickened sludge and dewatered sludge used in this study were collected from a wastewater treatment plant using the oxidation ditch process in Lanzhou. The seed sludge was obtained from thickened sludge after 12 days of AD under mesophilic conditions (35 ± 1 °C). The parameters of seed sludge and dewatered sludge are detailed (Table S1). Tetracycline was purchased from Shanghai Aladdin Biochemical Technology Co., Ltd.

The air was pumped into deionized water using a nanobubble generator (JXWNP-AL01, Jinxiang, China) at 22 ± 1 °C for 15 min (Wang et al., 2024). The inlet pressure during the generation process was maintained at 0.4–0.5 MPa. After generation, the suspension was

allowed to stand for 10 min, during which the solution gradually changed from milky white to transparent, indicating the formation of stable nanobubbles. The produced nanobubbles exhibited an average diameter of 135.8 nm with a concentration of approximately 5.37×10^8 bubbles mL^{-1} (Wang et al., 2025a). The detailed properties of Air-NBW are presented (Table S2).

2.2. Experimental setup

Experiments were conducted using the 2 L working volume digesters. According to the optimal dose reported in the previous study (seed sludge: dewatered sludge: Air-NBW = 1:6:4), each digester contained 160 g of seed sludge, 960 g of dewatered sludge, and 640 g of either Air-NBW or deionized water (serving as the control group) (Wang et al., 2025a). To better understand the fate of ARGs in sludge under antibiotic stress, different concentrations of tetracycline (0, 2, and 20 mg L^{-1}, corresponding to 0, 25, and 250 mg kg^{-1} dry weight) were added to the digesters. Anaerobic conditions inside the digesters were maintained by purging with nitrogen gas, after which all digesters were incubated in a water bath at 35 ± 1 °C for 38 days. The definitions of the experimental groups are shown in Table 1. All experimental groups were performed in triplicate.

2.3. Analytical methods

Periodic determinations of total solids (TS), volatile solids (VS), pH, total ammonia nitrogen (TAN), and soluble chemical oxygen demand (SCOD) were conducted on various sludge samples using standard methods (APHA, 2012). The calculation of free ammonia (FAN) was performed by specific formula, which relates temperature, TAN, and pH (Hansen et al., 1998). Methane content and volatile fatty acid (VFA) were detected by a gas chromatograph (Ruihong, SP-7890Plus, China). The biogas volume was measured by a wet gas flow meter (Alpha, LML-2, China). The elemental composition of different samples was measured by the elemental analyzer (Thermofisher, Flash 2000, USA). The pH, dissolved oxygen, conductivity, and temperature of NBW were monitored using the pH meter (Leici, PHS-3C, China), the dissolved oxygen meter (Sanxin, SX-620, China), the conductivity meter (Leici, DDSJ-307F, China), and the thermometer (Testo, Testo 720, China).

The sample was sieved through a 100-mesh sieve after being freeze-dried. McIlvaine-Na_2EDTA buffer and methanol were added to 0.5 g of the sample, following the extraction method described by Wang et al. (2019). After repeating the process three times, the supernatants were combined for solid-phase extraction. The solid-phase extraction column (HLB, 200 mg, 6 mL) was activated with 6 mL of methanol and 6 mL of ultrapure water (pH = 4) to keep the column moist. The extract was passed through the solid-phase extraction column at a flow rate of 1 mL min^{-1}. After the extract had fully passed through, the column was washed with 6 mL of ultrapure water, vacuum-dried for 20 min, and finally eluted with 9 mL of methanol. The eluted solution was collected and evaporated using a nitrogen blow-down apparatus until nearly dry. The volume was then adjusted to 1 mL with 10% methanol, filtered through a 0.22 μm nylon membrane, and stored for further analysis. The tetracycline residue concentrations were determined by liquid chromatography triple quadrupole tandem mass spectrometer (Waters, Xevo TQ-XS, USA) equipped with an electrospray ionization source, with detailed parameters provided in (Table S3 and S4). Measurements of total antioxidant capacity (*T*-AOC), superoxide dismutase (SOD), catalase (CAT), and ATP synthase activities in the AD system were performed following the method described by Yang et al. (2022).

Table 1
Definition of experimental groups.

Group	NBW addition	Tetracycline concentration
C0	No	0 mg L^{-1}
C2	No	2 mg L^{-1}
C20	No	20 mg L^{-1}
A0	Yes	0 mg L^{-1}
A2	Yes	2 mg L^{-1}
A20	Yes	20 mg L^{-1}

2.4. Metagenomic sequencing

To further investigate the impact of tetracycline on the functional potential of microorganisms during AD, metagenomic sequencing of sludge samples was conducted at different stages (Days 0, 4, 17, and 38) using the Illumina NovaSeq 6000 platform, with a sequencing mode of PE150. The original data obtained from sequencing were subjected to quality checks, including the removal of adapters and low-quality reads, thereby obtaining more accurate species annotation and functional annotation results (Xiang et al., 2024). Subsequently, de novo assembly was performed separately for each sample. The assembled contigs were subjected to CDS prediction, and short segment contig sequences were filtered out. Non-redundant Unigene sets were generated via sequence clustering. The Unigenes were compared with the NR_meta database to obtain species classification information at various levels. Then, the obtained data information was predicted or annotated according to the KEGG and the MEGs database (Peng et al., 2025). ARG identification and quantification were performed using the ARGs-OAP 3.0 pipeline, which integrates the SARG v3.0 database, with stringent alignment criteria (E-value: 1e-10; identity: 80%; query cover: 80%). To eliminate biases from sequencing depth and microbial biomass, raw ARG abundances were corrected by the pipeline's dynamic parameter k and further normalized to the 16S rRNA gene copy number (annotated via ARGs-OAP 3.0's count_16s module), and the final ARG abundance was expressed as ARG copies per 16S rRNA gene copy (Peng et al., 2025). This approach ensures the normalization of ARG abundance across all samples, allowing for reliable comparison of ARG dynamics without bias from sequencing depth or microbial biomass. The risk ranking of ARGs uses the Risk Ranking of SARG V3.0 database (Zhang et al., 2021). The potential ecological risk score (ERS) and human health risk score (HHRS) associated with ARGs in sludge samples were evaluated using MetaCompare 2.0 (Rumi et al., 2024). All original sequencing data have been submitted to the NCBI database, with the accession number PRJNA1369098.

2.5. Statistical analysis

SPSS version 27.0 was utilized for data analysis, and Origin 2022 was used to draw figures. Methane production was fitted using a modified Gompertz formula. Spearman's correlation was utilized for correlation analysis.

3. Results and discussion

3.1. Digestion performance and tetracycline degradation

NBW improved the hydrolysis performance and methane production of AD (Fig. 1). On day 4, Compared with the control groups (C0, C2, and C20), the VFA of A0, A2, and, A20 increased by 15.3% ($p < 0.05$), 13.3% ($p < 0.05$), and 10.7%, respectively (Fig. 1**a**), and the SCOD increased by 6.0% ($p < 0.05$), 4.7%, and 4.4%, respectively (Fig. S1). NBW may enhance microbial activity by accelerating electron transfer between microorganisms (Hou et al., 2024), thereby promoting the hydrolysis of organic matter. On day 38, the VFA increased with the elevation of tetracycline, with propionic acid accounting for the largest proportion. This might be because tetracycline affects microbial activity by inhibiting protein synthesis and altering cell membrane permeability, leading to the slow accumulation of VFA (Wang et al., 2025b). In addition, tetracycline may induce propionic acid accumulation by increasing the abundance of propionic acid-producing bacteria and reducing

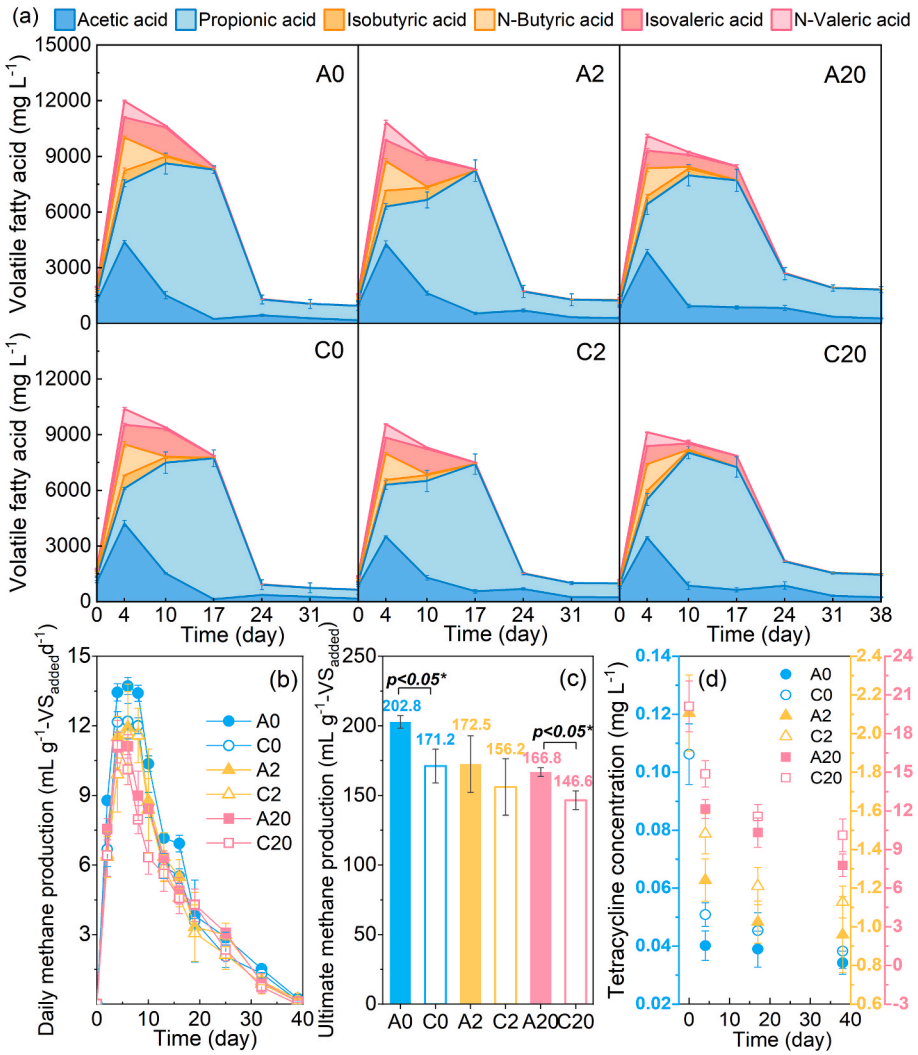


Fig. 1. Volatile fatty acid (a), daily methane production (b), ultimate methane production (c), and tetracycline concentrations (d) during anaerobic digestion.

propionate utilization efficiency (Wang et al., 2025b).

Compared with C0, the ultimate methane production of C2 and C20 decreased by 8.8% and 14.4%, respectively (Fig. 1**c**). This may be due to tetracycline primarily inhibiting methane production through indirect pathways such as system acidification, and the reduction in methane production exhibited a positive correlation with tetracycline concentration (Wang et al., 2025b). The maximum methane production rate (R_m) gradually decreased with increasing tetracycline, while NBW increased the R_m in all experimental groups (Table S5). Compared with the control groups (C0, C2, and C20), the ultimate methane production of A0, A2, and, A20 increased by 18.5% ($p < 0.05$), 10.4%, and 13.8% ($p < 0.05$), respectively (Fig. 1**c**). This increase in methane production may be attributed to the combined effects of NBW on sludge digestion performance and tetracycline stress mitigation. Owing to the large specific surface area and unique contact angle of nanobubbles, the contact area between microorganisms and substrates may increase, which enhances the biochemical reaction rate and promotes methane production (Chuenchart et al., 2021). The addition of NBW may optimize the mass transfer conditions of sludge AD and provide a more favorable microenvironment for the growth of methanogens (Hou et al., 2024). In addition, the increase in methane production may also be partly associated with the mitigation of tetracycline-induced stress on anaerobic microorganisms (Hou et al., 2024; Pan et al., 2019; Wang et al., 2025b). Therefore, the observed increase in methane yield is likely the result of the synergistic effects of improved substrate conversion and alleviated antibiotic stress.

Tetracycline is primarily removed through biodegradation, adsorption, and oxidative decomposition in AD (Yan et al., 2024). After AD, the tetracycline removal rates of C0, C2, and C20 reached 63.9%, 46.4%, and 49.8%, respectively (Fig. 1**d**). This might be because the hydrolases secreted by resistant hydrolytic bacteria may break the chemical bonds (e.g., ester bonds and peptide bonds) of tetracycline molecules, initially decomposing them into small-molecule intermediates (Zhong et al., 2022). Once these compounds are adsorbed onto the surface of microbial cells or enter the cells, intracellular oxidoreductases and lyases further catabolize them into carbon dioxide, water, organic acids, and other substances, which are eventually utilized by microorganisms (Lan et al., 2024; Zhao et al., 2024). Among the control groups, C0 exhibited the highest tetracycline removal rate, indicating that antibiotic stress inhibited the removal effect of AD on tetracycline. Biodegradation may be the dominant pathway for tetracycline removal during AD.

After AD, the tetracycline removal rates of A0, A2 and A20 were 67.7%, 54.4% and 61.4% respectively, further improving the tetracycline removal efficiency of AD (Fig. 1**d**). Hydroxyl radicals and superoxide radicals generated during nanobubbles collapse possess strong oxidizing properties, which could facilitate the stepwise transformation of tetracycline through demethylation, decarbonylation, hydrolysis, ozonation, decarboxylation, and alcohol oxidation (Chuenchart et al., 2021; Koundle et al., 2024). These transformation products may subsequently undergo further degradation and mineralization (Koundle

et al., 2024). Therefore, in addition to promoting microbial metabolic activity, the enhanced tetracycline removal observed in the NBW groups may also be associated with the oxidative processes related to reactive oxygen species (ROS) in the AD system (Chuenchart et al., 2021). Among NBW groups, the tetracycline reduction in A20 was greater than that in A2. This might be because in A2, the relatively low tetracycline toxicity had little impact on microbial activity, limiting the "mitigative effect" of NBW and resulting in a more moderate promotion of tetracycline removal (Hou et al., 2024; Wang et al., 2025b).

3.2. Microbial communities and methane metabolic pathways

Pseudomonadota, *Bacteroidota*, *Bacillota*, *Chloroflexota*, and *Actinomycetota* were the dominant phyla in the AD system, playing a predominant role during the hydrolysis and acidification stage (Fig. 2**a**). Initially, *Bacteroidota* exhibited the highest relative abundance. As AD progressed, *Pseudomonadota* gradually became the dominant phylum. This might be attributed to its strong tetracycline resistance mechanisms and ability to degrade tetracycline (Yang et al., 2024). *Bacillota* plays a vital role in the hydrolysis of polysaccharides and proteins (Wang et al., 2025b). Furthermore, *Bacillota* exhibits strong resistance to antibiotic stress and can effectively degrade tetracycline (Shen et al., 2020). Therefore, under the selective pressure of tetracycline, a bacterial community pattern has gradually formed, characterized by the proliferation of antibiotic-resistant phyla (*Pseudomonadota* and *Bacillota*) and the inhibition of sensitive phyla (*Bacteroidota*, *Chloroflexota*, and *Actinomycetota*) (Fig. 2**a**). Notably, compared with the control groups, the NBW groups exhibited a higher abundance of *Bacteroidota* and *Bacillota*. This might be due to the improvement of the hydrolytic bacterial environment by NBW, which has stimulated the activity of their metabolic enzymes (Wang et al., 2024).

Methanoculleus, *Methanothrix*, *Methanospirillum*, *Methanosarcina*, and *Methanofollis* were the dominant archaeal genera in the AD system, playing a predominant role during the methanogenesis stage (Fig. 2**b**). Initially, *Methanothrix* showed the highest relative abundance, with the acetoclastic methanogenesis pathway showing higher expression than the hydrogenotrophic methanogenesis pathway. The hydrogenotrophic methanogenesis pathway dominats during the late stage of AD. This might be because tetracycline can continuously reduce the relative abundance of *Methanothrix* by inhibiting upstream acetate production (Wang et al., 2025b). Moreover, as a typical hydrogenotrophic methanogen with high tolerance to antibiotics, *Methanoculleus* can adapt to the

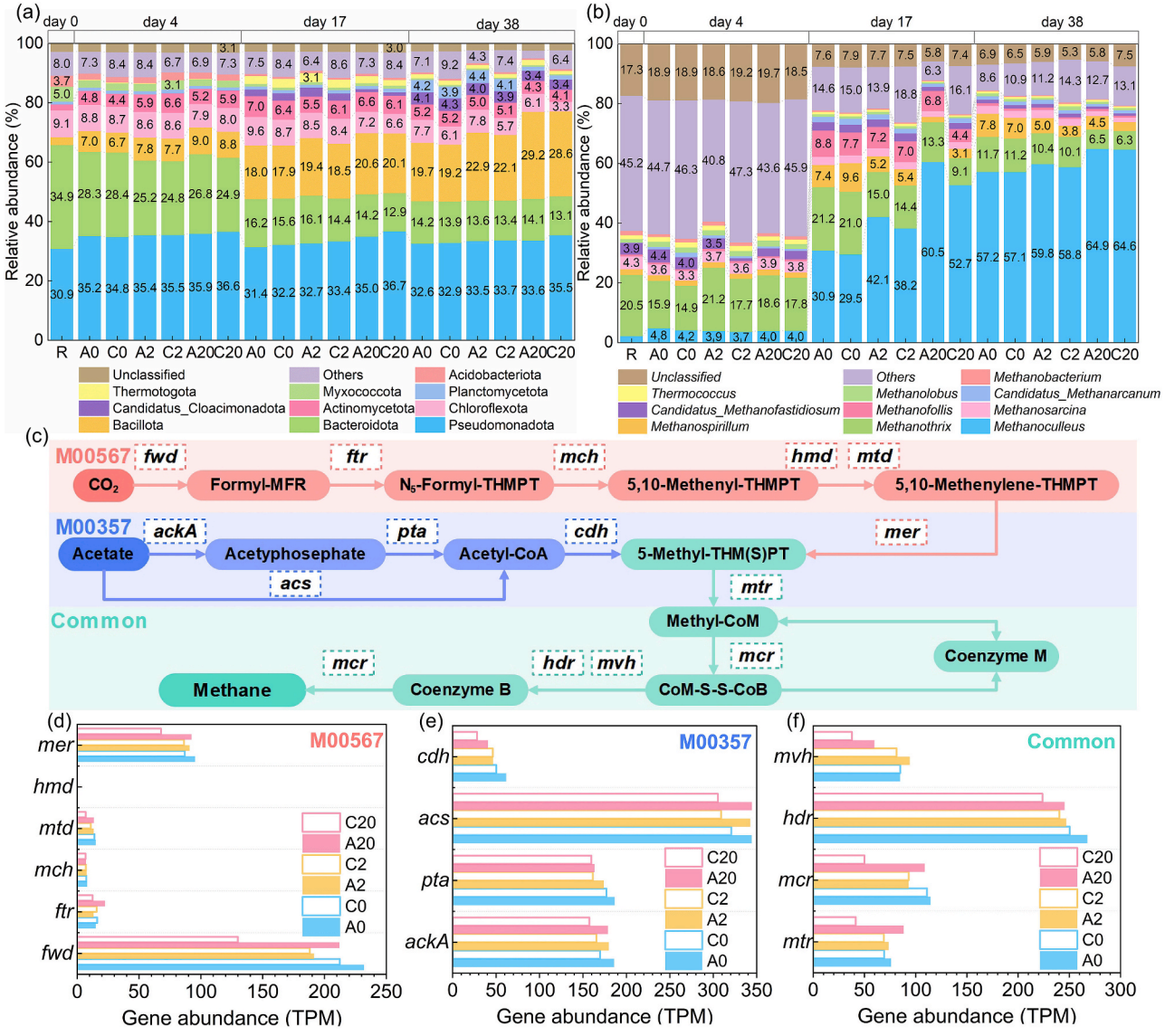


Fig. 2. Relative abundance of bacteria at the phylum level (a) and archaea at the genus level (b). The major methane metabolic pathways (hydrogenotrophic: M00567; acetoclastic: M00357) (c), and abundance of methanogenesis-related functional genes in different pathways on day 38 (d-f).

environmental fluctuations caused by tetracycline (Chen et al., 2024). After tetracycline inhibited some sensitive bacteria, hydrogen accumulation may occur in the system, providing *Methanoculleus* with sufficient substrates and leading to a significant increase in its relative abundance, thus becoming the dominant contributor to methanogenesis (Wang et al., 2025b). Notably, different types of methanogens exhibited improved relative abundances after NBW addition. This might be because NBW enhances mass transfer efficiency, enabling more efficient delivery of nutrients to the vicinity of methanogens, which facilitates the contact and utilization of nutrients by methanogens (Hou et al., 2024). Principal component analysis showed that digestion time exerted a more significant effect on microbial communities (Fig. S2).

The hydrogenotrophic methanogenesis pathway (M00567) and the acetoclastic methanogenesis pathway (M00357) are the two primary pathways of methane metabolism (Ma et al., 2024). In the hydrogenotrophic methanogenesis pathway, increased tetracycline downregulated the *fwd*, *ftr*, *mch*, *mtd*, and *mer* genes, inhibiting the bioconversion of carbon dioxide to 5-Methyl-THM(S)PT (Fig. 2**c and 2d**). The downregulation of the *ackA*, *pta*, *acs*, and *cdh* genes similarly indicated the inhibition of tetracycline on acetate conversion in the acetoclastic methanogenesis pathway (Fig. 2**c and 2e**). Compared with the control groups, most functional genes were significantly upregulated following NBW supplementation, demonstrating that NBW may promote organic matter conversion and methane production (Wang et al., 2024). Based on the highly expressed *fwd*, *acs*, and *hdr* genes across different pathways, the gene upregulation in A0 and A20 was significantly greater than that in A2, which further enhanced the bioconversion of 5-Methyl-THM(S)PT to methane. This may be due to the stable microbial functional community structure and high activity in A0, allowing the beneficial effects of NBW to be fully exerted and resulting in a significant improvement in methane production (Hou et al., 2021). In A20, NBW exhibited a more pronounced advantage in alleviating inhibition under high tetracycline, which may lead to a more significant recovery of methanogenic capacity (Fig. 1**d**).

3.3. Changes in total antibiotic resistance genes

ARGs are critical genetic contaminants in AD that pose potential environmental risks via microbial horizontal dissemination (Pang et al., 2025). As AD proceeded, the abundance of ARGs in C0 maintained a continuous downward trend until the end of AD (Fig. 3**a**). These results indicate that AD contributes to reducing the abundance of ARGs in sludge. MGEs serve as the core vectors for the horizontal gene transfer of ARGs and also function as critical mediators that facilitate the dissemination of ARGs among diverse microorganisms (Zhang et al., 2026). In the late stage of AD, tetracycline increased the abundances of MGEs (Fig. 3**b**), which may have promoted ARG dissemination and thereby contributed to ARG enrichment (Luo et al., 2020). ARGs were significantly enriched in C20, with the gene abundance exceeding that in the initial sludge on day 38 (Fig. 3**a**). This might be due to the fact that although AD contributes to the removal of ARGs, the addition of tetracycline still inhibits this process, which was corroborated by the previous study (Bai et al., 2019). After AD, compared with C0, the addition of tetracycline increased the abundance of tetracycline resistance genes (Fig. 3**a**). Among all tetracycline-added groups, the C20 exhibited the highest abundance of tetracycline resistance genes, which was consistent with the trend of total ARG abundance in each group on day 38. Furthermore, positive correlations were observed between tetracycline concentration and both total ARG abundance and tetracycline resistance genes abundance (Fig. 3**c and 3d**). This result indicates that the increase in tetracycline levels can promote the accumulation of ARGs in the AD system.

Compared with the control groups (C0, C2, and C20), NBW reduced the total abundance of ARGs by 1.8%, 4.0%, and 14.3%, respectively, and decreased the abundance of tetracycline resistance genes by 5.3%, −0.4%, and 13.9%, respectively (Fig. 3**a**). This is likely because high

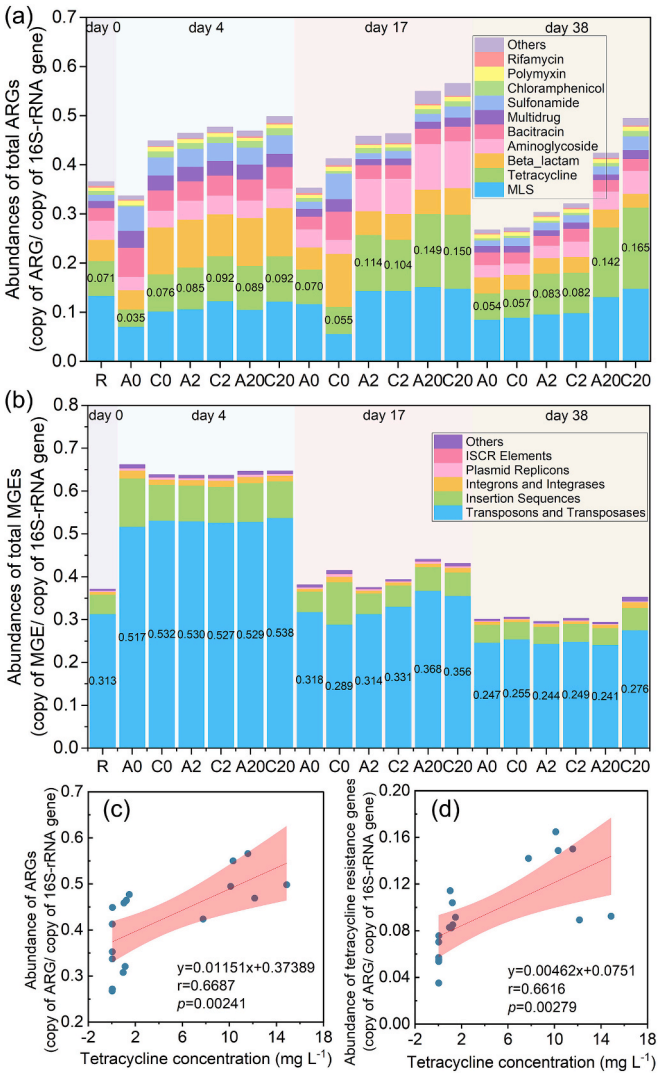


Fig. 3. Abundance of total antibiotic resistance genes (a) and total mobile genetic elements (b) during anaerobic digestion and linear relationships between tetracycline concentration and abundance of ARGs (c) or tetracycline resistance genes (d).

tetracycline concentrations directly enhance the selective advantage of resistant microorganisms, increase their survival and reproduction, and simultaneously increase the frequency of horizontal dissemination of ARGs among microorganisms, which may lead to a greater reduction effect of NBW on gene abundance (Huang et al., 2024). Notably, compared with the corresponding control groups, the abundance of ARGs was effectively reduced in A0 and A20, whereas a slight enrichment of ARGs was observed in A2. This phenomenon may be attributed to the combined effects of the species-specific response of bacteria and the differential variation of transposons and transposases (Maladan et al., 2024; Wang et al., 2025b). The tetracycline concentration applied in A2 corresponds to a relatively low or sub-inhibitory antibiotic level, which may exert selective pressure on microbial communities without strongly inhibiting microbial growth (Andersson and Hughes, 2014).

Specifically, as the core host of tetracycline resistance genes (e.g., *tet (36)*), the relative abundance of *Bacillota* exhibited a concentration-dependent increase in the control groups (Fig. 2**a**). Compared with the corresponding control groups, the magnitude of increase in the relative abundance of *Bacillota* in A2 was higher than that in A0 and A20 after the addition of NBW. This highest relative increase in A2 indicates that NBW might positively promote the growth and proliferation of *Bacillota*-

resistant subpopulations enriched under low tetracycline stress, thereby leading to a concomitant increase in the abundance of tetracycline resistance genes they harbor. Such selective enrichment may also be associated with microbial stress-response regulation triggered by low antibiotic exposure, which can facilitate the persistence or expression of resistance determinants in bacterial populations (Davies et al., 2006; Ma et al., 2024). In addition, the improvement of substrate availability and microbial metabolic conditions by NBW may unintentionally provide more favorable growth conditions for resistant subpopulations already adapted to low tetracycline stress, thereby partially counteracting the ARG-attenuating potential of NBW in this specific scenario (Andersson and Hughes, 2014). Furthermore, compared with the corresponding control groups, the abundances of transposons and transposases in A2 decreased by 2.0%, which was much lower than those in A20 (12.7%) and A0 (3.1%) (Fig. 3**b**). This indicated that the capacity of transposons and transposases to mediate ARG transfer was not significantly inhibited in A2. Under relatively mild antibiotic stress, microbial metabolic activity and DNA replication processes may remain active, which can further facilitate the persistence and potential dissemination of ARGs mediated by MGEs (Peng et al., 2025). Therefore, the proliferation of *Bacillota* resistant subpopulations and the retained transposition activity of MGEs synergistically contributed to the slight enrichment of tetracycline resistance genes in A2. Taken together, the result observed in A2 may reflect a competitive relationship between the alleviating effect of NBW and the selective enrichment induced by low-level tetracycline, in which the latter temporarily predominates under this condition (Kraemer et al., 2019).

3.4. Changes in tetracycline resistance genes

Among the top 10 tetracycline resistance genes with the highest abundance (Fig. S3), *tet(M)* poses a direct threat to human health as Class I risk ARGs, while *tet(W)*, *tetX2*, and *tetA(P)* may pose potential future risks as Class II risk ARGs (Zhang et al., 2024). At the end of AD, the abundances of most genes in C0 exhibited a decreasing trend. In addition, high tetracycline stress affected the abundance of tetracycline resistance genes during sludge AD. Except for *tetX2*, *tet(C)*, and *tetA(P)*, the abundances of the remaining tetracycline resistance genes in C20 significantly increased after AD. Most tetracycline resistance genes were positively correlated with resistant bacteria, *Pseudomonas* (Fig. S4). There are three main types of tetracycline resistance genes, efflux pump genes (*tet(C)*, *tetA(P)*), ribosomal protection protein genes (*tet(W)*, *tet(32)*, *tet(M)*, *tet(36)*, *tet(Q)*, *tet(O)*, *tet(T)*), and enzymatic modification genes (*tetX2*) (Wu et al., 2010). High tetracycline stress may inhibit the expression of efflux pump genes due to abnormal transporter expression and impaired energy metabolism (Hong et al., 2014). Ribosomal protection protein genes are mainly present in Gram-positive bacteria, and these microorganisms possess growth advantages under high tetracycline stress, which may lead to a subsequent increase in the abundance of ribosomal protection protein genes (Blake et al., 2025). Oxidoreductases encoded by enzymatic modification genes rely on oxygen or specific electron donors for their function (Krueger and Williams, 2005). Under anaerobic conditions, enzymatic activity is restricted, and high tetracycline may further exacerbate their metabolic pressure, thereby inhibiting the expression of these genes.

NBW reduced the abundance of most tetracycline resistance genes (Fig. S3). The A20 exhibited the most pronounced changes, particularly among Class I risk ARGs, where the abundances of *tet(M)* decreased by 22.2%, compared with C20. *Tet(W)* was the most abundant tetracycline resistance gene detected. Compared with the control groups, its abundance in the NBW groups decreased by 3.7–13.5%. This might be due to NBW promoting the degradation of tetracycline, reducing the selection pressure of tetracycline, which leads to a decrease in the abundance of tetracycline resistance genes (Fig. 1**d**) (Koundle et al., 2024).

3.5. Functional gene expression and oxidative stress levels

Biofilm formation serves as a self-protective mechanism for microorganisms to cope with environmental stress (Xiang et al., 2024). After AD, compared with C0, the abundance of functional genes associated with biofilm formation in C2 and C20 was gradually upregulated (Fig. 4**a**). This might be because the survival pressure imposed by tetracycline on the microorganisms triggered their stress response mechanisms, resulting in the upregulation of functional genes associated with biofilm formation, in order to construct the biofilm (Peng et al., 2025; Wang et al., 2025b). However, biofilm formation can induce microbial aggregation, thereby promoting the horizontal dissemination of ARGs (Peng et al., 2025). Compared with the control groups, the abundance of functional genes associated with biofilm formation in the NBW groups was downregulated by 2.1–11.7 TPM. Therefore, the addition of NBW may reduce the risk of ARG dissemination. The type IV secretion system (T4SS) enables the transcellular transport of effector proteins and serves as a core functional system that mediates the conjugative transfer of MGEs (Xiang et al., 2024). The virB4 and virB8 are the most abundant functional genes associated with T4SS (Fig. 4**e**). Based on their abundance changes, the increased tetracycline concentrations upregulated the expression of functional genes associated with T4SS, thereby facilitating the conjugative transfer of MGEs. Compared with the control groups, the addition of NBW downregulated the abundance of functional genes associated with T4SS by 6.1–17.8 TPM (Fig. 4**d**). This indicates that NBW could inhibit the conjugative transfer of MGEs under tetracycline stress.

Tetracycline stress can induce ROS accumulation and oxidative stress in microorganisms (Zheng et al., 2023). Compared with C0, the abundance of functional genes associated with antioxidant defense was upregulated in C2 and C20 (Fig. 4**b**). This might be because, in response to oxidative stress, microorganisms upregulate antioxidant functional genes to enhance the activities of antioxidant enzymes, thereby reducing ROS levels (Xiang et al., 2024). DNA damage may be triggered by excessive oxidative stress, with the microbial DNA repair system being activated in response (Lu et al., 2018). Compared with C0, the abundance of functional genes associated with DNA repair and recombination was upregulated in C2 (Fig. 4**c**). However, the abundance of these functional genes downregulated with increasing tetracycline concentration. This may result from the high toxicity of high-concentration tetracycline to microorganisms, which leads to partial cellular damage or death and thus reduces the activity of the DNA repair system (Xiang et al., 2024). At the end of AD, compared with the control groups, the addition of NBW led to an increase of 10.0–22.0 TPM and 20.0–111.9 TPM in the abundance of functional genes associated with antioxidant defense and DNA repair and recombination, which contributed to improving the systemic antioxidant capacity.

To better understand the impact of NBW on the antioxidant system of microorganisms, the activities of related enzymes were determined. *T*-AOC serves as a comprehensive indicator reflecting the overall antioxidant capacity (Yang et al., 2022). On day 4, *T*-AOC in the NBW groups was lower than that in their corresponding control groups (Fig. 4**f**). As AD progressed, *T*-AOC in the NBW groups gradually exceeded that of the control groups. SOD and CAT are key antioxidant enzymes, and increased activities of these enzymes may reduce ROS-induced damage to microbial cells (Li et al., 2021). The trend of SOD activity was consistent with that of *T*-AOC (Fig. 4**h**). This may be because in the early stage of AD, superoxide anion generated during NBW collapse directly reacts with indigenous SOD in sludge, leading to its rapid consumption (Chuenchart et al., 2021; Li et al., 2022). However, the initiation of antioxidant functional gene expression in microorganisms requires time, which may delay the timely replenishment of SOD and other antioxidant substances, resulting in the temporarily lower contents of SOD and *T*-AOC in the NBW groups compared with the control groups (Fig. 4**f and 4h**). At this time, trace ROS produced by nanobubbles has begun to activate the antioxidant defense system of microorganisms. On day 38,

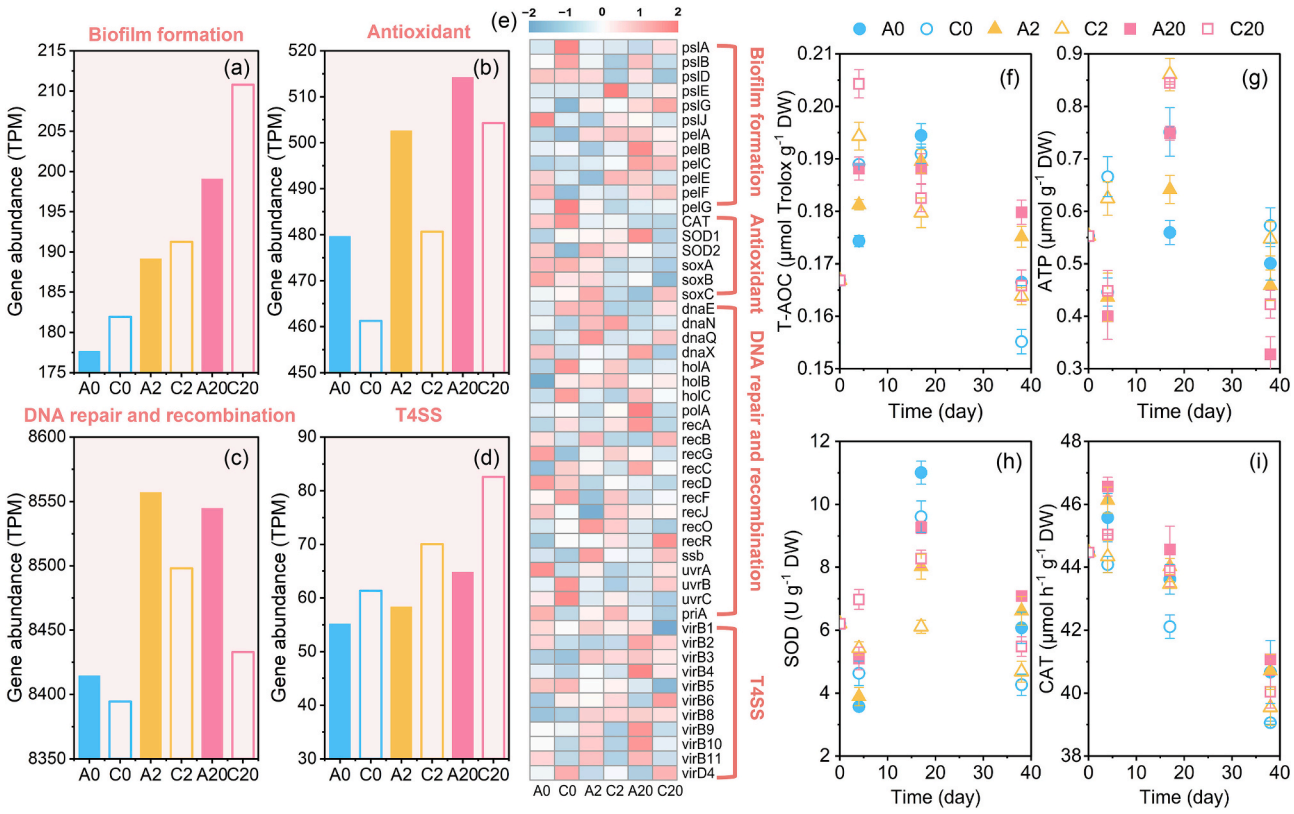


Fig. 4. The expression levels of functional genes associated with biofilm formation (a), antioxidant defense (b), DNA repair and recombination (c), and T4SS (d) on day 38, and the individual gene expression heatmap of samples (e). Changes in *T*-AOC (f), ATP (g), SOD (h), and CAT (i) contents during anaerobic digestion.

SOD and *T*-AOC contents in the NBW groups were increased by 29.3–42.3% and 6.9–8.4% compared with the control groups. This might be because NBW promoted organic matter degradation, reduced the accumulation of inhibitory metabolites such as VFAs, and optimized the microbial metabolic environment, helping to allocate more energy and substances to the synthesis of antioxidants (Chuenchart et al., 2021). Additionally, NBW upregulated the abundance of functional genes associated with antioxidant defense, which may lead to an increase in the activities of antioxidant enzymes (Fig. 4**b**).

CAT catalyzes the decomposition of hydrogen peroxide into water and oxygen, thereby functioning in the subsequent step of ROS scavenging after SOD converts superoxide anion into hydrogen peroxide (Li et al., 2022). Therefore, CAT did not need to undergo a consumption phase in the early stage of AD, leading to its consistently higher activity in the NBW groups than in the control groups throughout AD (Fig. 4**i**). Compared with the control groups, ATP contents in the NBW groups decreased by 12.6–22.5% after AD (Fig. 4**g**). This may be due to the fact that NBW promotes the degradation and transformation of organic matter, enhances the metabolic activity of microorganisms and the rate of ATP turnover, and enables the generated ATP to be consumed more quickly for substrate utilization, cell damage repair, and methane production and other life activities (Hou et al., 2024).

3.6. Potential action mechanisms of nanobubble water and final digestate risks

In summary, NBW treatment downregulated the abundance of functional genes associated with T4SS (Fig. 4**d**), upregulated the abundance of functional genes associated with antioxidant defense (Fig. 4**b**), and enhanced the activity of antioxidant enzymes in microorganisms (Fig. 4**h and 4i**). Based on these findings, this study suggests that NBW may enhance the activities of antioxidant enzymes by upregulating the abundance of functional genes associated with antioxidant defense, thereby alleviating oxidative stress (Chuenchart et al., 2021; Hou et al., 2024). This reduction in oxidative stress could help maintain microbial cell stability and function, thereby mitigating the negative impact of antibiotics (Yang et al., 2022). Secondly, NBW may inhibit the conjugative transfer pathway of MGEs by downregulating the abundance of functional genes associated with T4SS, resulting in a decrease in the activities of MGEs (Zhang et al., 2026). These pathways may collectively reduce the horizontal dissemination of ARGs, contributing to the observed decrease in ARG abundance. At present, this phenomenon is considered a potential mechanistic explanation, and further targeted experiments are needed to provide deeper insights into the underlying mechanisms.

AD reduced the ERS and HHRS from 7.03 and 0.57 to 6.68 and 0.41 (Fig. 5**b and 5c**). However, the ERS and HHRS of C2 and C20 increased by 0.18–0.58 and 0.08–0.19, respectively, compared with C0. It can be seen that tetracycline increased the potential ecological and human health risks of ARGs in sludge AD. Notably, the change rate of HHRS was higher than that of ERS in C2 and C20, suggesting that under the same tetracycline concentration conditions, the change in potential human health risks is greater than the change in potential ecological risks. This phenomenon was probably attributed to the fact that high-risk ARGs closely associated with human health are more prone to enrichment under tetracycline stress (Zhang et al., 2021). Meanwhile, these ARGs may exhibit higher environmental exposure risks and more direct transmission pathways (e.g., entering the food chain through sludge land application or via water contact) (Ben et al., 2019).

Compared with the corresponding control groups, the addition of NBW reduced ERS (Fig. 5**b**), indicating that NBW reduced the potential ecological risks of ARGs under tetracycline stress to some extent. Therefore, the introduction of NBW may provide a novel practical strategy for alleviating the environmental risk caused by ARGs in sludge AD. Although the present study demonstrated the effectiveness of NBW in enhancing tetracycline degradation and ARG reduction in laboratory-

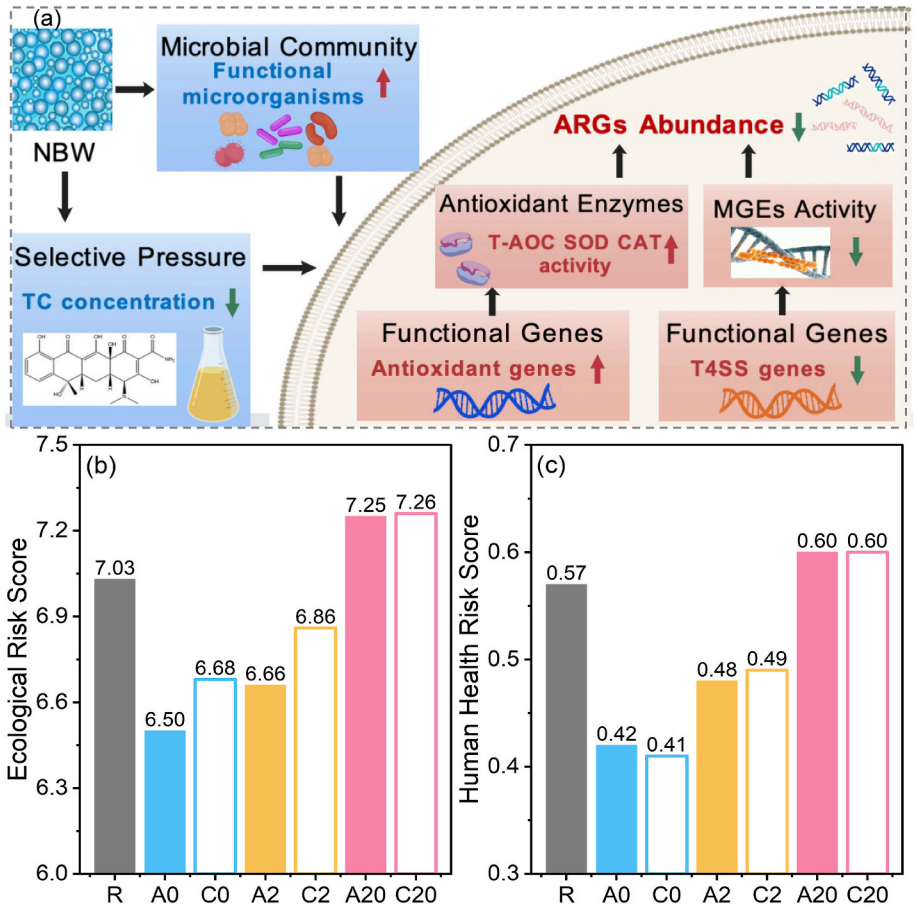


Fig. 5. Schematic diagram of the potential mechanisms of antibiotic resistance gene abundance reduction induced by nanobubble water during anaerobic digestion (a). The antibiotic resistance genes ecological risk scores (b) and human health scores (c) of the final digestate.

scale AD, its practical application in full-scale systems still requires cautious evaluation (Ansari et al., 2025). As a liquid-phase additive, NBW may offer practical advantages in avoiding additional solid residues and simplifying sludge management (Chuenchart et al., 2021). However, large-scale implementation may still face challenges related to the energy demand of nanobubble generation, equipment investment, long-term operational stability, gas–liquid mass transfer efficiency, and treatment cost (Ansari et al., 2025).

4. Conclusion

NBW promoted methane production, with the promoting effect decreasing as tetracycline concentration increased. In sludge AD, NBW promoted tetracycline degradation, and reduced the abundance of ARGs by enhancing the antioxidant capacity of the system and mitigating ARG horizontal dissemination potential. These findings indicate that, at the laboratory scale, NBW is a promising liquid-phase additive for reducing ARG-related environmental risks in AD systems. Future studies should focus on long-term operation, scale-up performance, and the integration of NBW with other treatment technologies.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used ChatGPT in order to improve the readability and language of the manuscript. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

CRediT authorship contribution statement

Ziying Xu: Writing – review & editing, Writing – original draft, Validation, Methodology, Investigation, Formal analysis, Data curation. **Tianfeng Wang:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition. **Yuxuan Zuo:** Validation, Investigation, Formal analysis. **Yuxin Zhang:** Investigation, Formal analysis. **Yuqian Zhang:** Investigation, Formal analysis. **Dan Luo:** Investigation, Formal analysis. **Yonggang Wang:** Investigation, Formal analysis. **Liqing Xin:** Resources, Investigation, Funding acquisition. **Wei Peng:** Resources, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.biortech.2026.134703.

Data availability

Data will be made available on request.

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